

DEQ project update

1 Executive summary

This project develops a reversible deep-equilibrium (RevDEQ) language model for the Parameter Golf budget (16 MB artifact, 600 s on 8×H100, scored by validation bits-per-byte, BPB). Unlike a fixed stack or loop of layers, it computes its output as the *fixed point* of one shared block — so its central question is convergence: does the iteration actually settle? We track the spectral radius $\rho(J_F) < 1$ as the convergence criterion and add a consistency signal that makes the model *learn* to converge. In the current model the solve moved from divergent ($\rho(J_F) = 1.30$) to contractive ($\rho(J_F) = 0.77$) while BPB improved monotonically to 1.4629 — stability and accuracy advancing together. A reversible backward pass keeps activation memory independent of solve depth, so deep iterations train on a single 48 GB GPU. Next, we will rigorously justify a convergence guarantee at no cost to performance and benchmark against the recurrent-depth literature; we seek the advisor’s read on the direction.

2 Problem and goal

The goal is a language model formulated as a genuine fixed-point solve — with measurable, provable convergence and aggressive parameter reuse — that is competitive under the Parameter Golf budget (artifact ≤ 16 MB, training ≤ 600 s on 8×H100, FineWeb, vocab 1024, scored by BPB). Parameter reuse (“recurrent depth”) is a leading route to small models, but most implementations loop a block a fixed number of times with no guarantee that the iteration converges; we aim to close that gap and contribute a convergence framework that transfers across iterative architectures. All numbers here are controlled ~ 1000 -step development runs (one variable at a time); the public leaderboard’s ~ 1.06 BPB is a long-term target, not a matched comparison (different tokenizer, $\sim 5\times$ compute, and an evaluation/quantization stack we omit).

3 Approach

The model computes its representation as the fixed point of one shared, weight-tied block. For input embeddings x_0 ,

$$z^* = T_\theta(z^*, x_0), \quad T_\theta(z, x_0) = \bar{B} \odot \text{RMSNorm}(x_0) + \Delta_\theta(z, x_0),$$

where the $\bar{B}$ term injects the input and Δ_θ is the block update. Capacity comes from 16 experts in a dense *soft*-routed mixture: every expert sees every token through smooth softmax routing (hard top- K dispatch would break the reversible solve), and each owns a low-rank multi-head latent attention (MLA, rank 64) and MLP (rank 96) path. The model totals 13.9 M parameters (model dim 768, sequence length 2048, tied embeddings); “12 layers” denotes the *maximum solver depth* K , not twelve distinct parameter sets.

Forward solve. The fixed point is reached by a two-state damped (Parcae) iteration. With a learned per-dimension retention $\bar{A} \in [\varepsilon, 1)$ and blend $\beta = 1 - \bar{A}$,

$$\begin{aligned} y_{k+1} &= \bar{A} \odot y_k + \beta \odot T_\theta(z_k), \\ z_{k+1} &= \bar{A} \odot z_k + \beta \odot T_\theta(y_{k+1}), \end{aligned}$$

run for K steps from an initial state, with $z_K \approx z^*$ (we suppress the fixed input x_0 inside T_θ). The damping $\bar{A}$ keeps each step a small, stable move toward equilibrium.

Reversible backward and VRAM efficiency. The update is *exactly invertible*: given (y_{k+1}, z_{k+1}) the previous state is recovered by

$$\begin{aligned} z_k &= (z_{k+1} - \beta \odot T_\theta(y_{k+1})) \oslash \bar{A}, \\ y_k &= (y_{k+1} - \beta \odot T_\theta(z_k)) \oslash \bar{A}, \end{aligned}$$

where $\oslash$ denotes per-dimension division. A naive K -step unroll must *store* all K intermediate states for backpropagation, so activation memory grows as $O(K)$. Instead, the backward pass *reconstructs* each state by running these inverses from the final state — activation memory is therefore $O(1)$ in K , independent of how deep the solve runs. The reciprocals require $\bar{A}$ to stay off zero (a reversibility floor $\varepsilon = 0.1$) and are evaluated in fp64 so the reconstruction matches the forward to machine precision; gradients then flow through a short truncated window (depth 3). The payoff is concrete: deep solves (K up to 128) train in ≈ 38 GB — within a single 48 GB (L40S-class) GPU — where an $O(K)$ unroll of the same depth would not fit. Reversibility is what makes a low-resource machine sufficient for this research.

4 Convergence and evaluation

A fixed-point model is only meaningful if the iteration actually settles. For the solve to reach a fixed point the map must be locally contractive, and the standard certificate is that the spectral radius of its Jacobian stays below one ($\rho(J_F) < 1$). We therefore track $\rho(J_F)$ as a first-class metric — not the cruder operator-norm bound, which would flag as “divergent” solves that in fact converge — and we make the model *achieve* convergence with a consistency signal that ties shallow and deeper solves to the same answer, a target it can only meet if its iteration contracts. Convergence is thus learned, not hand-tuned; the results below show it improving *together* with accuracy.

What we measure. Table 1 lists every tracked metric, why it matters, and its current value. Beyond the score, the metrics certify the two things the architecture relies on: that the solve *converges* ($\rho(J_F)$ and the residual), and that the 16 experts stay *healthy* (balanced, distinct, and actually used).

Metric	What it measures	Why we measure it	Value
<i>Objective and convergence</i>			
BPB (primary)	bits/byte on full validation, post-quant	the scored objective — the bottom line	1.4629
$\rho(J_F)$	spectral radius of the iteration Jacobian	certifies the fixed point converges (the model's premise)	0.77
$\sigma_{\max}(J_F)$	operator norm of the Jacobian	shows the operator-norm bound is too conservative (reported, not gated)	≈ 14.5
conv. residual	relative change at the deepest K	empirical check that the solve has settled (< 0.05 healthy)	0.011
K -extrap. drift	BPB from $K=16$ to $K=128$	confirms the solve holds at test depths beyond training	+0.6 mBPB
<i>Expert / routing health (16 experts; attn/MLP pairs)</i>			
load balance (CV)	spread of token mass over experts	detects imbalance or collapse onto a few experts	0.32 / 0.10
min share	smallest per-expert usage	flags dead experts (wasted capacity)	3.9% / 5.6%
per-token entropy	how concentrated each token's routing is	measures specialization (tokens using few experts)	3.25
utilization entropy	usage spread across the batch (max 3.47)	measures whether <i>all</i> experts are used	3.44
output orthogonality	cosine between expert outputs	detects redundant (collinear) experts	0.05 / 0.23
iteration variation	routing change across solver steps	whether the iterative depth is actually used	0.001–0.005
<i>Budget</i>			
artifact size	total bytes (weights + code)	must fit the 16 MB cap	7.9 MB
quantization gap	BPB pre- vs. post-int6 quant	cost of compressing the model into the artifact	+26 mBPB

Table 1: Metrics tracked every evaluation. Paired values (e.g. 0.31/0.09) are the attention and MLP routers.

Behaviour across solver depth. Table 2 complements Table 1 by sweeping fixed solver depths K , now including the expert-health metrics. The model has learned a genuine equilibrium, not a memorized step count: BPB is essentially flat once $K \geq 16$ and the convergence residual falls monotonically to ≈ 0.011 , while the shallow solves ($K=4, 8$) are still settling. $\rho(J_F)$ here is a power-iteration estimate and is noisy at individual K ; the dependable evidence is the residual and the settled deep- K value (≈ 0.77), while $\sigma_{\max}$ stays $\gg 1$ throughout — which is exactly why it is the wrong gate. The expert-health columns barely move with K — routing is insensitive to eval depth, the same signal behind the effective-depth-1 problem. Hardening the $\rho(J_F)$ estimator is the first next step (Section 6).

K	BPB	resid.	$\rho(J_F)$	$\sigma_{\max}$	$CV_{a/m}$	ortho $_{a/m}$	H_{tok}
4	1.6235	0.304	1.08	13.3	0.34/0.10	0.05/0.23	3.24
8	1.4745	0.076	0.79	14.0	0.32/0.10	0.05/0.23	3.25
16	1.4716	0.013	1.25	15.0	0.32/0.10	0.05/0.23	3.25
24	1.4720	0.011	0.87	14.6	0.32/0.10	0.05/0.23	3.25
32	1.4721	0.011	1.02	14.7	0.32/0.10	0.05/0.23	3.25
64	1.4722	0.011	0.77	14.3	0.32/0.10	0.05/0.23	3.25
128	1.4722	0.011	0.77	14.5	0.32/0.10	0.05/0.23	3.25

Table 2: Comprehensive K -sweep of the current model (fixed- K post-quant diagnostic; the headline full-validation BPB is 1.4629). Columns span both convergence (BPB, conv. residual, $\rho(J_F)$, $\sigma_{\max}$) and expert health (load-balance CV, output orthogonality, per-token entropy H_{tok} ; “ a/m ” = attention / MLP routers). BPB is flat for $K \geq 16$ and the residual falls monotonically; $\rho(J_F)$ is a power-iteration estimate, noisy at individual K (> 1 at points) but settling to ≈ 0.77 deep, while $\sigma_{\max} \gg 1$ throughout (the over-restrictive gate). The expert metrics are essentially K -invariant — routing does not vary with eval depth, consistent with the effective-depth-1 observation.

The convergence arc. The decisive result is that BPB and convergence improved *together* (Table 3): the consistency signal first pulled $\rho(J_F)$ from a divergent 1.30 to 0.92 at near-flat BPB, then its recursive multi- K form pushed $\rho(J_F)$ to 0.77 *and* delivered the largest BPB gain.

Stage	bpb	$\rho(J_F)$	Change
baseline	1.4718	$\approx 1.30^\dagger$	no convergence signal; solve not yet contractive
+ consistency signal	1.4716	0.92	convergence, not BPB, was the win
+ recursive multi- K	1.4629	0.77	−8.7 mBPB; current model

Table 3: Controlled ~ 1000 -step development runs. † Without the convergence signal the solve is not contractive: $\rho(J_F) \approx 1.30$ at training onset; the signal then trains it below 1 — to 0.92, then 0.77.

Health and budget. The expert metrics are healthy — all 16 experts active and well utilized — and the artifact uses only 50% of budget, leaving room to grow capacity.

5 Related work (comparison set)

We will benchmark against the recurrent-depth and equilibrium-model literature. The principal comparators:

- **DEQ / TorchDEQ.** The original deep-equilibrium models: solve for a fixed point instead of stacking layers. Our backbone is a reversible DEQ variant.
- **RevDEQ.** Reversible DEQ with exact gradients and fewer function evaluations; the method closest to ours, which we extend with dense expert routing and explicit $\rho(J_F)$ diagnostics.
- **Universal Transformer.** The seminal model that applies one shared block recurrently through depth.

- **Parcae.** Stable looped LMs via per-dimension damping and input injection; we adopt its damping in our solver.
- **Huginn.** A large latent-recurrent LM with variable test-time compute.
- **MoEUT.** Adds sparse mixture-of-experts capacity to recover the expressivity lost under block sharing.
- **Mixture-of-Recursions.** Routes tokens to adaptive recursion depths — a dynamic-depth comparator.

6 Open problems and next steps

Open problems.

- **Effective depth ≈ 1 / weak expert specialization.** Routing barely changes across iterations (variation ≈ 0.001 – 0.005), so experts do not specialize by depth and the model under-uses its iterative computation.
- **K -extrapolation.** Trained shallow ($\mathbb{E}[K] \approx 24$) but must hold at $K=128$; eased, not eliminated, by the consistency signal.
- **Quantization gap.** Fast proxy 1.4369 $\rightarrow$ full post-quant 1.4629 (+26 mBPB).

Next steps.

1. *Rigorously justify a convergence metric* that guarantees asymptotic fixed-point convergence with no worse performance, and design the training so this convergence is reached by the time training itself converges.
2. *Drive expert specialization across depth.* Push the routing to differentiate from one solver iteration to the next, so experts specialize by depth and the iterative computation is genuinely used (today’s effective depth is ≈ 1).
3. *Enrich the Parcae damping/mixing with manifold-constrained hyper-connections* (DeepSeek mHC): replace the single damped residual stream with several parallel streams mixed by a doubly-stochastic, magnitude-preserving matrix. That magnitude preservation aligns naturally with our reversibility and $\rho(J_F) < 1$ requirements, and the extra streams are a candidate source of the differentiated state that step 2 needs.
4. *Compare under a matched (baseline) setting* against the related work in Section 5.
5. *Optional.* Sparsity and similar architectural changes, pursued only as throughput optimizations, not part of the core objective.

7 Summary

The convergence bet is working: we adopted the spectral radius $\rho(J_F) < 1$ as the criterion, measure it directly rather than a conservative proxy, and made the model *learn* convergence — the solve is now contractive ($\rho(J_F) = 0.77$) and BPB improved as a consequence, within budget and within a single 48 GB GPU thanks to the reversible backward. The next step is to make that convergence guarantee rigorous at no performance cost and benchmark it against the recurrent-depth literature. We welcome the advisor’s read on the direction.